dff9-W4111-Spring-2026 - HW3-Setup.ipynb

Part 0 — Introduction

Overview

HW3, HW4 and H5 require the use of MongoDB and Neo4j NoSQL databases. Students installed MySQL locally on their personal computers. Students will use "cloud versions" of MongoDB and Neo4j. Specifically,

- MongoDB Atlas
- Neo4j Aura

This notebook walks students through the process of registering, using and connecting to MongoDB Atlas and Neo4j Aura.

Submission Instructions

There is no submission for this task. Students should complete the task as soon as possible to prepare for HW3.

We will use the Ed Megathread to discuss the tasks and answer questions.

Part 1 — MongoDB

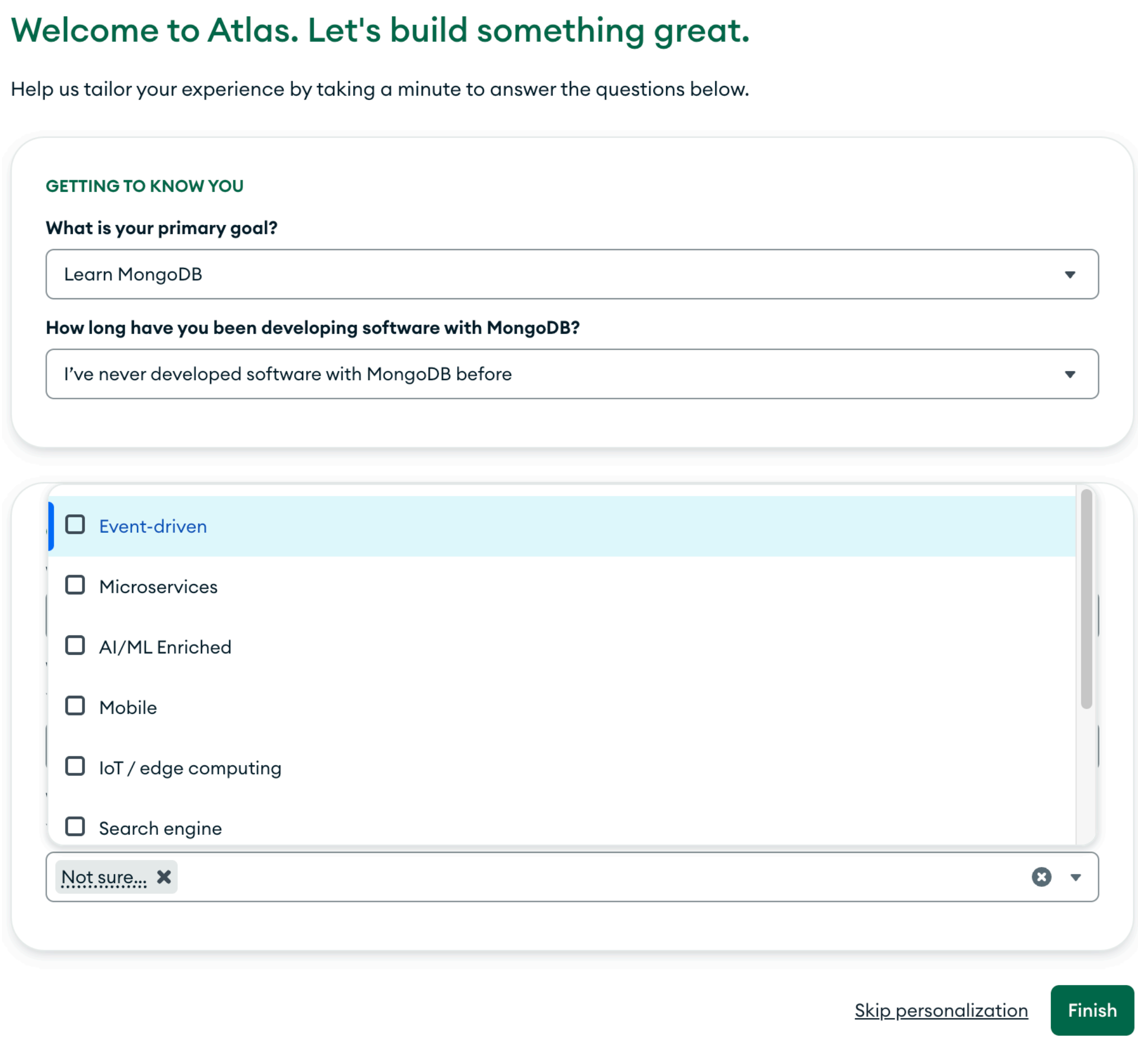
Register and Setup

1. Go to https://www.mongodb.com/products/platform/atlas-database and click on "Get Started."

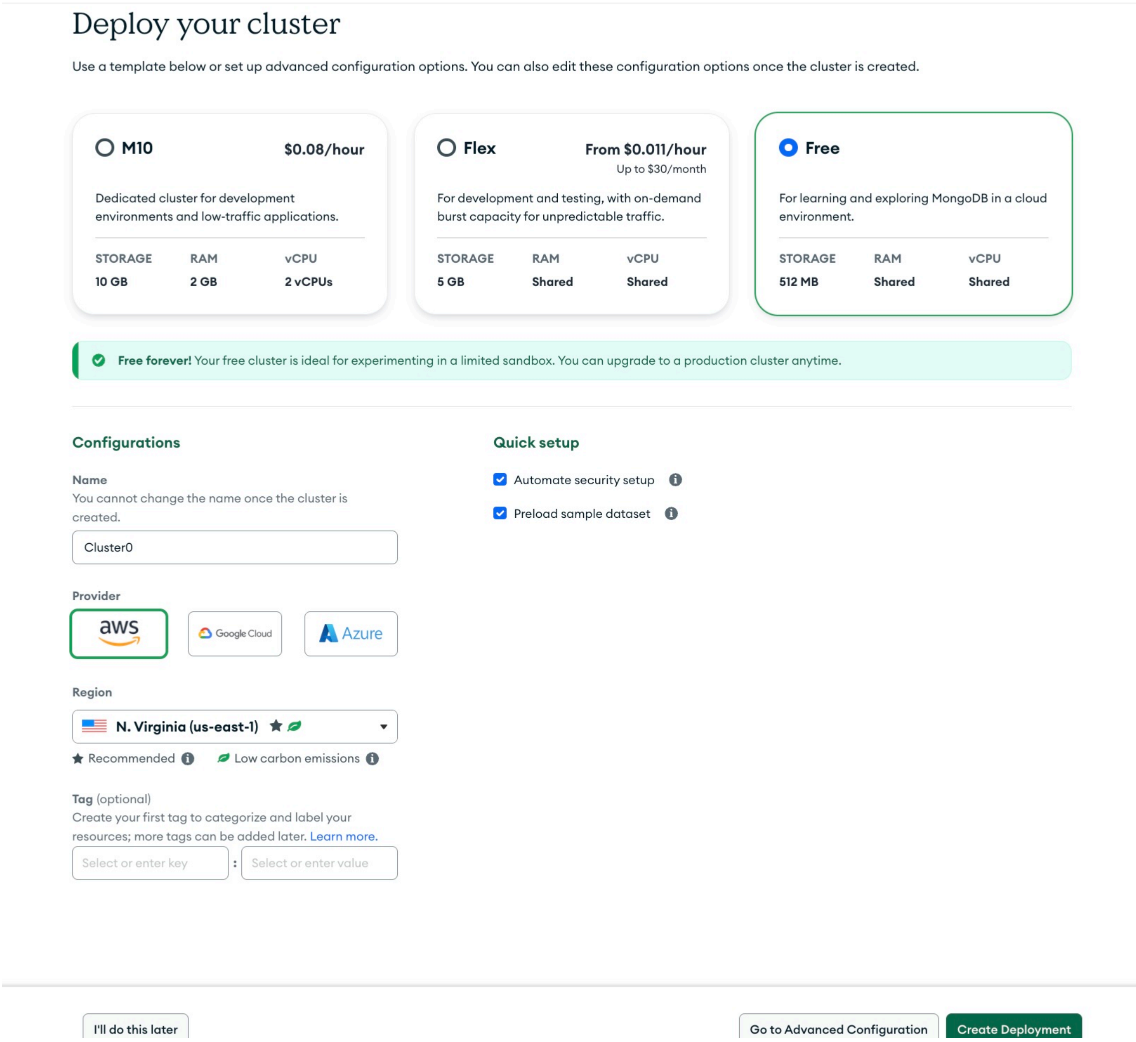
2. Click "Signup with Google" and use your Columbia ID or some other Gmail/Google account.

3. Accept the privacy policy and terms of service.

4. Answer the questions about tailoring the experience. The answers do not matter except for the programming language. For this question, select "Python." My answers look like



5. On the cluster creation screen make sure you select the free option. I chose AWS for my cloud provider and suggest you do the same. My screen looks like

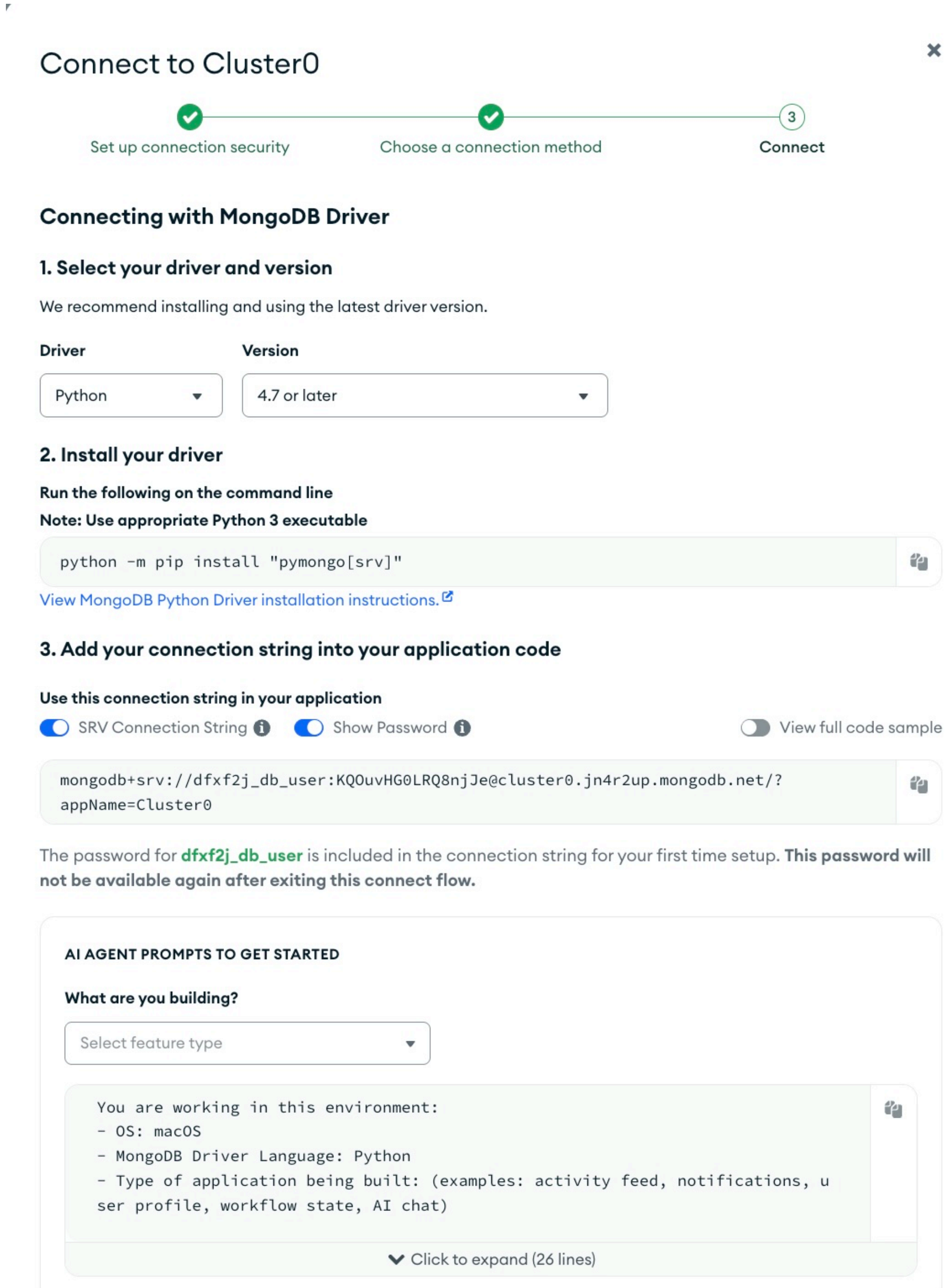


6. Select create cluster.

7. When prompted, create the database user. **Save the Username and Password**. You will need them for later. My values are in the cell below.

In [ ]:
```
mongodb_user="mb4882"
mongodb_pw="Alswo!3518?"
```

8. Choose connecting through a driver and select Python for the language. My selection looks like below. **Make a copy of the connection URL and put in the cell below.**



In [ ]:

```
# mongodb_url = "mongodb+srv://mb4882:Alswo!3518?@cluster0.xhp5mjw.mongodb.r
# mongodb_url = "mongodb+srv://mb4882:Alswo!%21518%3F@cluster0.xhp5mjw.mongo
```

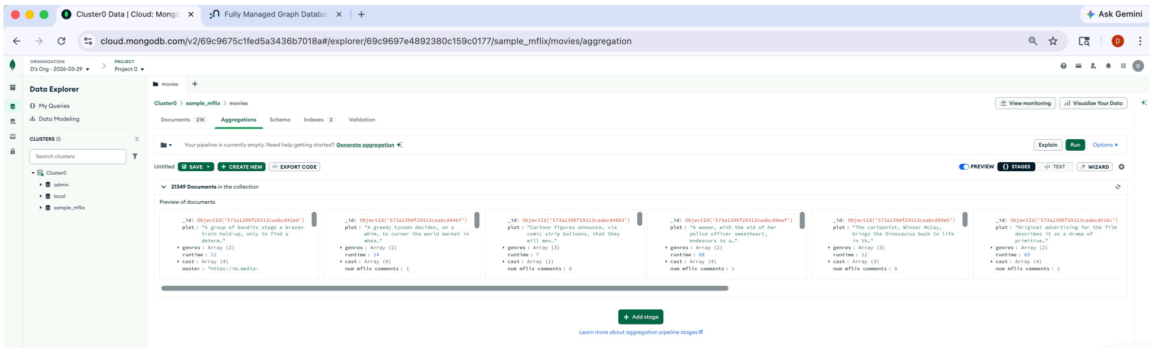
In [8]:

```
from urllib.parse import quote_plus

user = "mb4882"
password = "Alswo!3518?"
```

```
host = "cluster0.xhp5mjw.mongodb.net"
mongodb_url = f"mongodb+srv://{quote_plus(user)}:{quote_plus(password)}@{hos
```

9. Click on "Data Explorer" to explore your data. You should see a screen that looks something like.

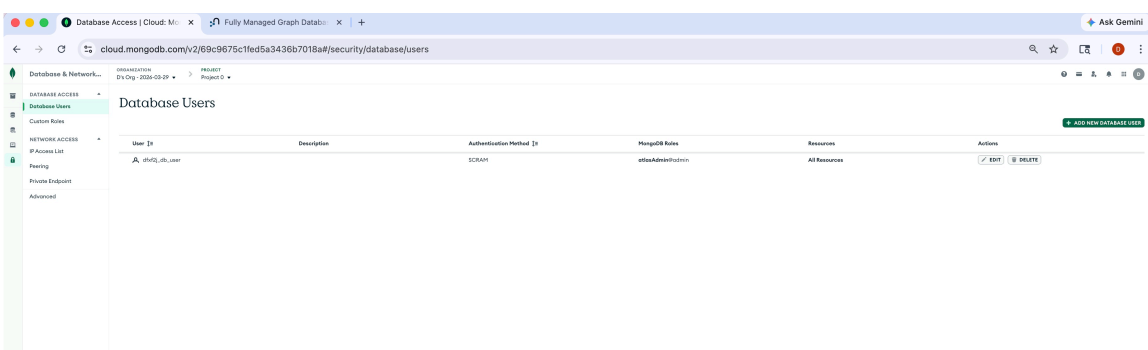


10. If you see this screen, you have completed the first part of setting up and connecting to MongoDB.

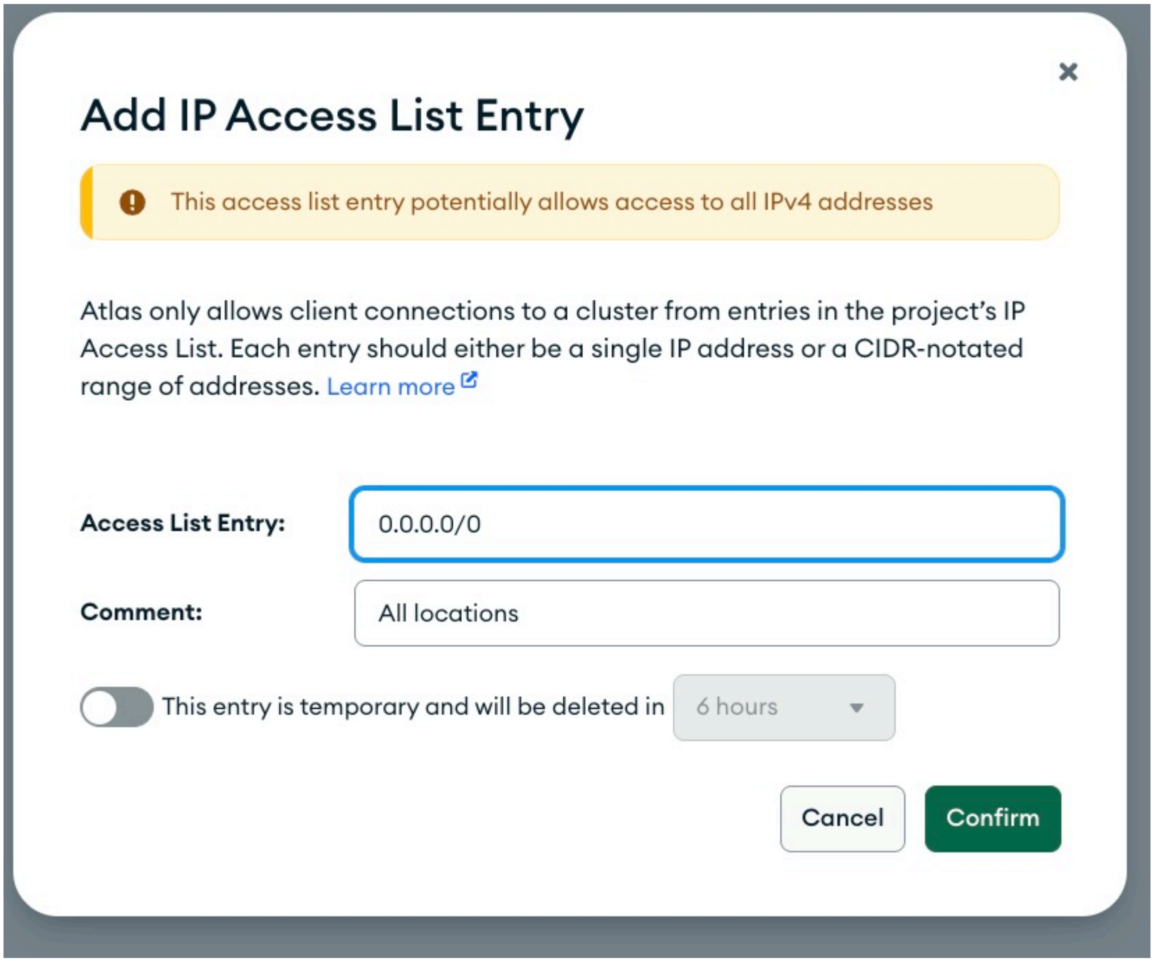
Connect from Jupyter Environment

1. Move your mouse over the leafs at the top of the browser window and the menu will expand. Choose "Database and Network Access".

2. After choosing, you will see a window that looks like.



3. Select "IP Access List." Choose "Add IP Address and enter 0.0.0.0/0. Your popup will look like below and click confirm.



4. Execute the following cell. Your output make look different. As long as you do not get any errors, you are all set.

In [1]:
```
%pip install "pymongo[srv]"
```

```
Requirement already satisfied: pymongo[srv] in /Users/minjae/miniconda3/envs/
database_hw3/lib/python3.10/site-packages (4.16.0)
WARNING: pymongo 4.16.0 does not provide the extra 'srv'
Requirement already satisfied: dnspython<3.0.0,>=2.6.1 in /Users/minjae/minic
onda3/envs/database_hw3/lib/python3.10/site-packages (from pymongo[srv]) (2.
8.0)
Note: you may need to restart the kernel to use updated packages.
```

5. Execute the following cell.

In [9]:
```
from pymongo.mongo_client import MongoClient
from pymongo.server_api import ServerApi

uri = mongodb_url

# Create a new client and connect to the server
client = MongoClient(uri, server_api=ServerApi('1'))

# Send a ping to confirm a successful connection
try:
```

```
    client.admin.command('ping')
    print("Pinged your deployment. You successfully connected to MongoDB!")
except Exception as e:
    print(e)
```

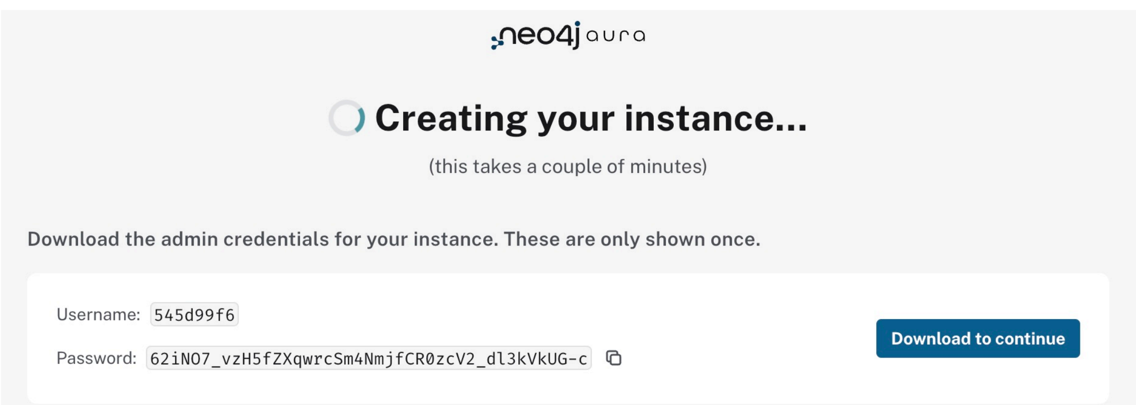
```
Pinged your deployment. You successfully connected to MongoDB!
```

6. If you get "Pinged your deployment. You successfully connected to MongoDB!", you are all setup with MongoDB Atlas.

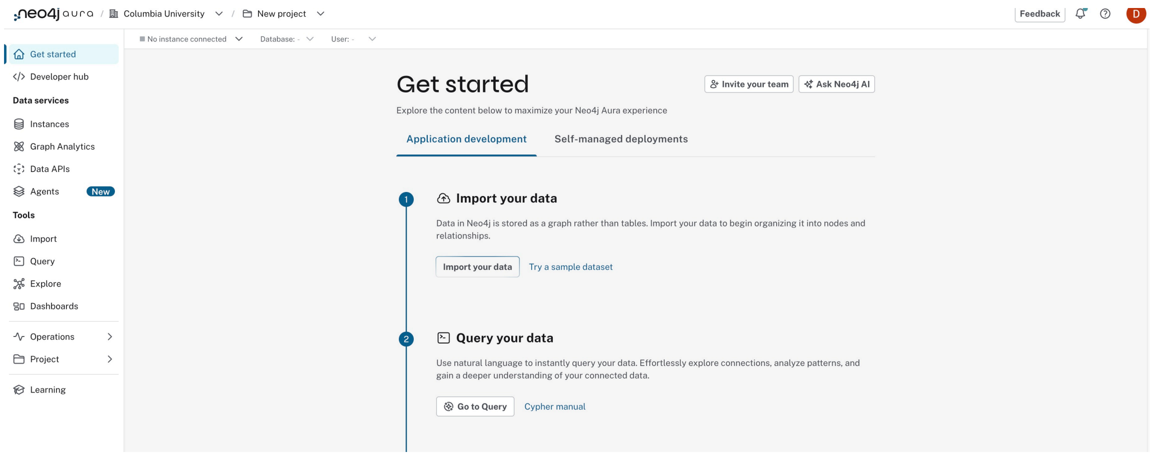
Part 2 — Neo4j

Register and Setup

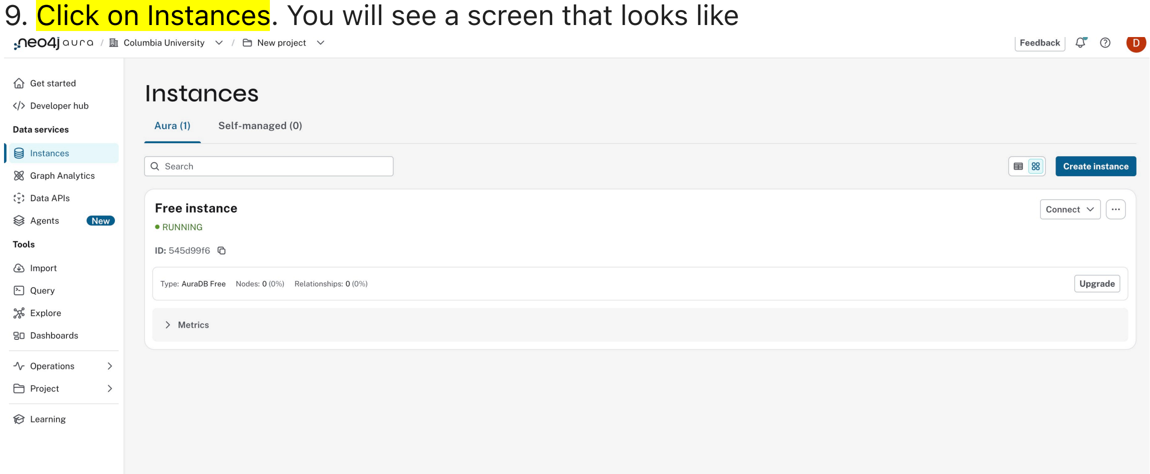
1. Go to https://neo4j.com/product/auradb/ and choose "Get Started Free."
2. Choose "Continue" with Google and click through the Google SSO Authorization.
3. Enter you name and "Columbia University" on the setup form.
4. On the "What are you looking to achieve with Neo4j?" select "Just exploring ... ..."
5. On the following popups, just select any answers. They are not important.
6. When prompted, click "Create Instance".
7. While Aura is creating your instance, you will see a popup that looks like the following. Select "Download to continue."



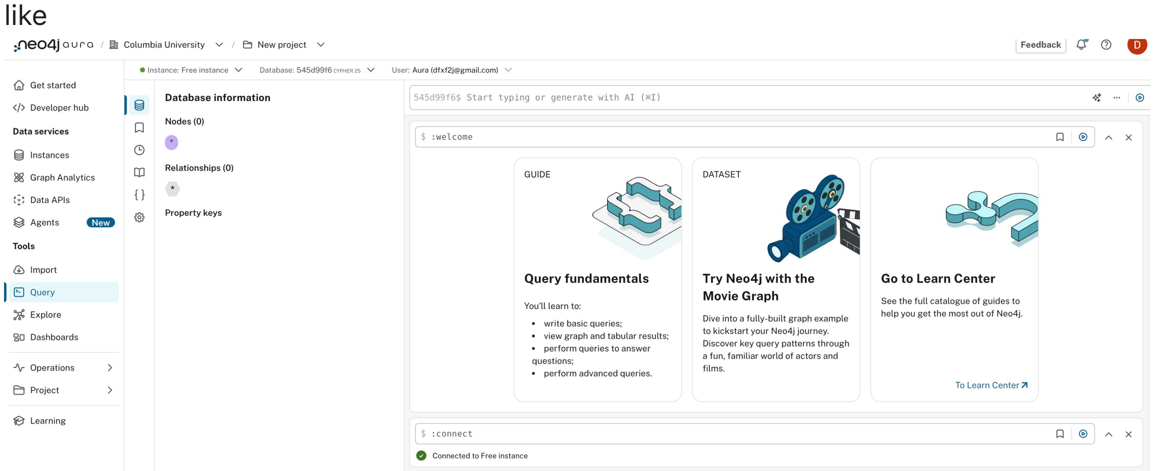
8. Once your instance is created, you will see a screen that looks like



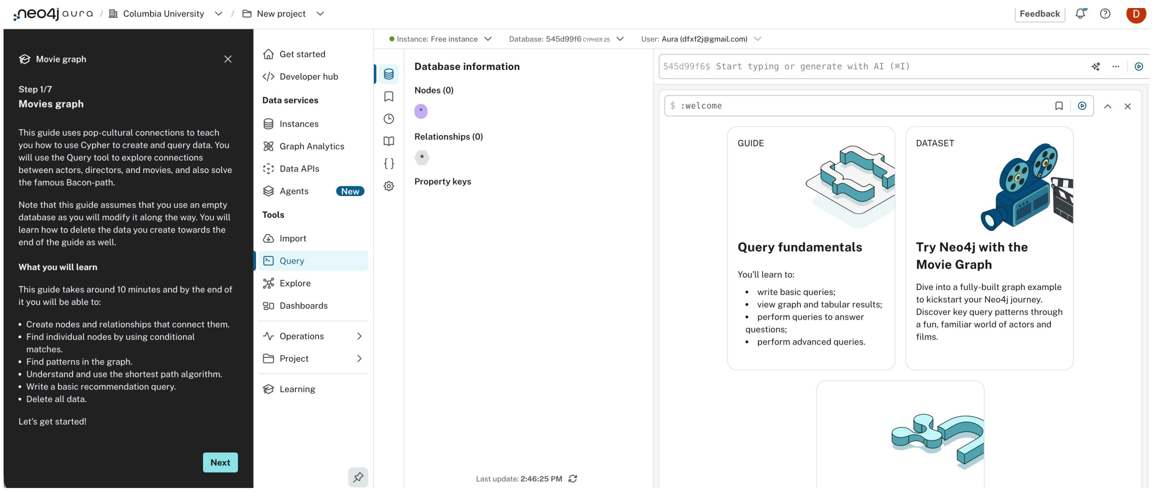
9. Click on Instances. You will see a screen that looks like



10. On the Connect pulldown, choose query. You will see a screen that looks something like



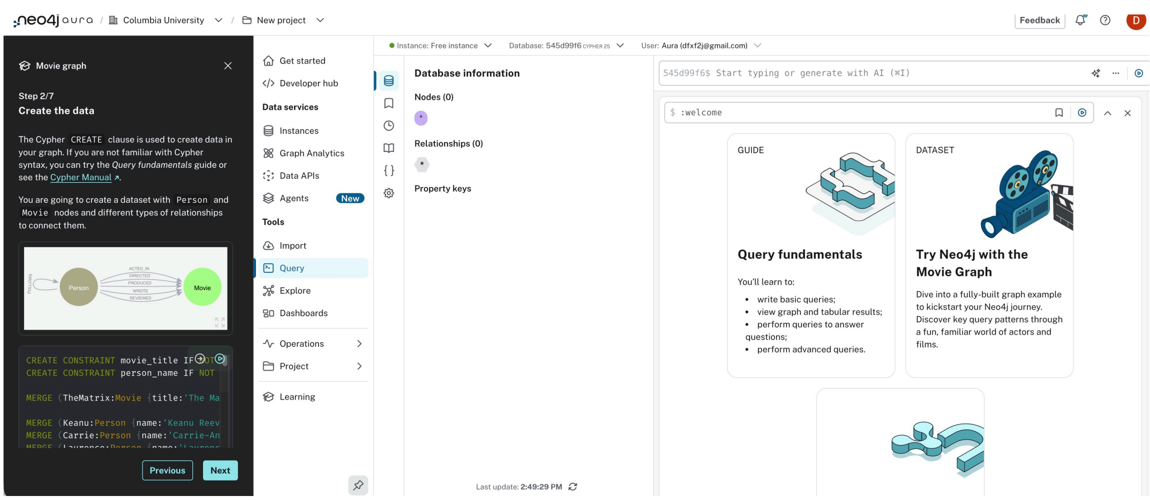
11. Choose "Try Neo4j with the Movie Graph." Your screen will look like



12. Click Next

Take the Tutorial

When you clock Next, you will start taking a simple tutorial. The first page will look like



You will take the simple tutorial. Note,

- The tutorial window is scrollable. So make sure you scroll through each page in the tutorial.
- When you see a cell with code, clock on the rightmost arrowhead at the top to execute the code.
- The results of the execution will appear on the right side of your window.

Test Connecting from Jupyter Notebook

Execute the following cell. Your result make look different from mine but you are fine as long as you do not get any errors.

In [10]:
```
%pip install neo4j
```

```
Collecting neo4j
  Downloading neo4j-6.1.0-py3-none-any.whl.metadata (5.3 kB)
Collecting pytz (from neo4j)
  Using cached pytz-2026.1.post1-py2.py3-none-any.whl.metadata (22 kB)
Downloading neo4j-6.1.0-py3-none-any.whl (325 kB)
Using cached pytz-2026.1.post1-py2.py3-none-any.whl (510 kB)
Installing collected packages: pytz, neo4j
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2/2 [neo4j]
Successfully installed neo4j-6.1.0 pytz-2026.1.post1
Note: you may need to restart the kernel to use updated packages.
```

In your downloads folder, you will see a file named something like `Neo4j-545d99f6-Created-2026-03-29.txt`. Open the file in a text editor, and copy and paste the text in the cell below.

1. Edit the cell to put quotes around the text strings.
2. Change "neo4j+s" to "Neo4j+ssc"

When you are done, your cell will look something like the one below. You can execute the cell.

In [ ]:
```
# # Wait 60 seconds before connecting using these details, or login to https
# NEO4J_URI="neo4j+ssc://545d99f6.databases.neo4j.io"
# NEO4J_USERNAME="545d99f6"
# NEO4J_PASSWORD="Some Password"
# NEO4J_DATABASE="545d99f6"
# AURA_INSTANCEID="545d99f6"
# AURA_INSTANCENAME="Free instance"
```

In [12]:
```
# Wait 60 seconds before connecting using these details, or login to https:/
NEO4J_URI="neo4j+ssc://d640b480.databases.neo4j.io"
NEO4J_USERNAME="d640b480"
NEO4J_PASSWORD="G1fJc6zyHCKMwKGTvP5qSfs2jwcmF3V4wOAd9hha_I0"
NEO4J_DATABASE="d640b480"
AURA_INSTANCEID="d640b480"
AURA_INSTANCENAME="Instance01"
```

Execute the following cell - Neo4j connection & sample query test

In [15]:
```
from neo4j import GraphDatabase

# URI examples: "neo4j://localhost", "neo4j+s://xxx.databases.neo4j.io"

URL = NEO4J_URI

AUTH = (NEO4J_USERNAME, NEO4J_PASSWORD)

with GraphDatabase.driver(NEO4J_URI, auth=AUTH) as driver:
    driver.verify_connectivity()

    records, summary, keys = driver.execute_query("""
```

```
        MATCH (p:Person {name: "Tom Hanks"})
        RETURN p
    """,
    database_=NEO4J_DATABASE,
)

# Loop through results and do something with them
for record in records:
    print(record.data())  # obtain record as dict

# Summary information
print("The query `{query}` returned {records_count} records in {time} ms.".f
    query=summary.query, records_count=len(records),
    time=summary.result_available_after
))
```

```
{'p': {'born': 1956, 'name': 'Tom Hanks'}}
The query `
        MATCH (p:Person {name: "Tom Hanks"})
        RETURN p
    ` returned 1 records in 12 ms.
```

If you do not get an error and your output is similar to mine, congratulations. You have successfully connected to Neo4j.

In [ ]: